Chapter four

1.1 Interpolation and Curve Fitting

Introduction

In this chapter, the problem of representing functions with a computer is discussed. Several different sub problems will be considered. They differ according to the type of function being represented, whether it is known at relatively few points or at many (or all) points. Efficient ways of representing date are fundamental to advancing the understanding of scientific problems. The representation choice (a polynomial, a spline function, and so on) also determines the nature of the theory. We begin with the case of polynomial, which is the oldest and simplest.

Many times, data is given only at discrete points such as $(x_0, y_0), (x_1, y_1), \ldots\ldots, (x_{n-1}, y_{n-1}), (x_n, y_n)$

So, how then does one find the value of y at any other value of x? Well, a continuous function $f(x)$ may be used to represent the $n+1$ data values with $f(x)$ passing through the $n+1$ points. Then one can find the value of y at any other value of x. This is called *interpolation*.

Curve fitting is the process of finding equations to approximate straight lines and curves that best fit given sets of data. Regression is the process of finding the dependent variable from some data of the independent variable. Regression can be linear (straight line) or curved (quadratic, cubic, etc.)

Data and Interpolating Functions

A function is said to **interpolate** a set of data points if it passes through those points.

Definition: The function $y = P(x)$ interpolates the data points $(x_1, y_1), \ldots, (x_n, y_n)$ if $P(x_i) = y_i$ for each $1 \le i \le n$.

Given a set of points (experimental data, table values, design stations, etc.) fit a curve f(x) to the points so that we can evaluate f(x) for any x of interest.

If x within data range —>**interpolating**

If x outside data range—>**extrapolating**

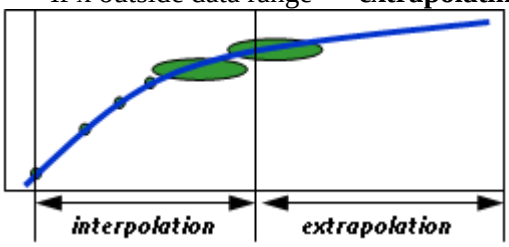


Figure 1

Plotting a smooth curve through discrete data points

Suppose we have a sequence of data points

In engineering and computer graphics, curve fitting techniques plays a very important role. In the most general sense, curve fitting involves the determination of a continuous function

y = f(x)

In fitting a curve to given data points, there are two general approaches. One is to have the approximating curve pass exactly through all of the known data points, which can be used when the data values are known to be very accurate.

The other one uses an approximating function, which graphs as a smooth curve having the general shape suggested by the distribution of the data, but not actually passing through all of them. For example, the plot of the *x* and *y* values shown insuggests a straight-line fit, using a function such as

y = a + bx.

Interpolation & Curve-fittingUsed for:

– Identification of underlying relationship (modeling/prediction)

– Interpolation (filling in the gaps)
– Extrapolation (predicting outside the range of the data)

Applications of Curve Fitting

- Reduce noise and smooth data
- Find mathematical relationship or function among variables. Use that function for further data processing such as
  - error compensation
  - velocity and acceleration calculation
- Estimate the variable value between data samples
- Estimate the variable value outside the data sample range

4.1 Polynomial Interpolation

Polynomials are the most common choice of interpolates relative to other choices such as a trigonometric and exponential series; because they are easy to:

- Evaluate
- Differentiate, and
- Integrate.

Polynomial interpolation involves finding a polynomial of order n that passes through the $n+1$ points. The method of Polynomial interpolation involves Lagrange's Method, Newton's Method and Neville's Method.

P is required to be a function; that is, each value x corresponds to a single y. This puts a restriction on the set of data points that can be interpolated – the x_i's must be all distinct in order for a function to pass through them. There is no such restriction on the y_i's.

To begin, we will look for an interpolating polynomial. We solve the following problem: we are given a table of $n+1$points (x_i, y_i)

x	x_0	x_1	x_2	$\cdots$	x_n	
y	y_0	y_1	y_2	$\cdots$	y_n	

And we seek a polynomial $P_n(x)$ of degree $\le n$ for which

$P_n(x_i) = y_i, (0 \le i \le n)$,

Theorem (Main Theorem of Polynomial interpolation)

If $x_0, x_1, \cdots, x_n$ are distinct real numbers, then for arbitrary values $y_0, y_1, \cdots, y_n$,

There is a unique polynomial $P_n(x)$ of degree $\le n$ such that

$P_n(x_i) = y_i, (0 \le i \le n)$.

Proof.For any polynomial $P_n(x)$ of degree$\le n$, we have the following form:

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n.$$

Then, to determine the polynomial $P_n(x)$ is to find the coefficient a_i's. We shall use the interpolation condition that

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + a_3 x_0^3 + \cdots + a_n x_0^n = y_0 \\ a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^3 + \cdots + a_n x_1^n = y_1 \\ a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^3 + \cdots + a_n x_2^n = y_2 \\ \cdots \\ a_0 + a_1 x_n + a_2 x_n^2 + a_3 x_n^3 + \cdots + a_n x_n^n = y_n \end{cases}$$

Note that, in equation the above $x_0, x_1, \cdots, x_n$, $y_0, y_1, \cdots, y_n$ are prescribed, therefore the unknowns are $(a_0, a_1, \cdots, a_n)^T$.

We have the following matrix form

$$\begin{pmatrix} 1 & x_0 & x_0^2 & x_0^3 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & x_2^3 & \cdots & x_2^n \\ 1 & x_3 & x_3^2 & x_3^3 & \cdots & x_3^n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \cdots & x_n^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix}.$$

Since the matrix in equation of above is a **Vander Monde matrix**, it is nonsingular if and only if $x_0, x_1, \cdots, x_n$ are distinct. This imply that the system exists a unique solution $(a_0, a_1, \cdots, a_n)^T$ if and only if $x_0, x_1, \cdots, x_n$ are distinct. Hence, there exists a unique polynomial $P_n(x)$ of degree $\le n$ if $x_0, x_1, \cdots, x_n$ are distinct.

4.1.1 Lagrange's Method

By the known function tables $(x_i, f(x_i)), (i = 0,1,\cdots,n)$ of $y = f(x)$, to seek $L_n(x) \in H_n$ which satisfies the interpolation conditions:

$f(x_i) = L_n(x_i), (i = 0,1,\cdots,n)$.

Obviously, the n-degree interpolation polynomial satisfied the conditions is

$$L_n(x) = \sum_{k=0}^{n} f(x_k) l_k(x) = f(x_i) \quad (i = 0,1,\cdots,n)$$

The interpolation polynomial $L_n(x)$ is also known as the Lagrange interpolation polynomial.

$$L_n(x) = \sum_{k=0}^{n} f(x_k) l_k(x) \quad \Rightarrow \quad P(x) = y_o L_o(x) + \cdots + y_n L_n(x) \text{ where}$$

$$l_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_i}{x_k - x_j}, (k = 0,1,\cdots,n) \quad \Rightarrow \quad L_k(x) = \frac{(x - x_0)(x - x_1)\cdots(x - x_{k-1})(x - x_{k+1})\cdots(x - x_n)}{(x_k - x_0)(x_k - x_1)\cdots(x_k - x_{k-1})(x_k - x_{k+1})\cdots(x_k - x_n)}$$

Example 1 - Given the following functional table 3.1:

x	2	2.5	4
$f(x)$	0.5	0.4	0.25

Find the second Lagrange interpolating polynomial.

Let $x_0 = 2,\ x_1 = 2.5,\ x_2 = 4$.

We determine the coefficient polynomials $L_0(x),\ L_1(x), L_2(x)$:

$$L_0(x) = \frac{(x-2.5)(x-4)}{(2-2.5)(2-4)} = (x-6.5)x+10,$$

$$L_1(x) = \frac{(x-2)(x-4)}{(2.5-2)(2.5-4)} = \frac{(-4x+24)x-32}{3},$$

and

$$L_2(x) = \frac{(x-2)(x-2.5)}{(4-2)(4-2.5)} = \frac{(x-4.5)x+5}{3}.$$

Since $f(x_0) = f(2) = 0.5$, $f(x_1) = f(2.5) = 0.4$ and $f(x_2) = f(4) = 0.25$, we have

$$P(x) = \sum_{k=0}^{2} f(x_k) L_k(x)$$

$$= 0.5\left((x-6.5)x+10\right) + 0.4\frac{(-4x+24)x-32}{3} + 0.25\frac{(x-4.5)x+5}{3}$$

$$= (0.05x-0.425)x+1.15$$

Example 2 - Find the Lagrange form of the interpolation polynomial $P_2(x)$ which interpolates the following data:

x	x_0	x_1	x_2	
y	y_0	y_1	y_2	

Purpose: seek a polynomial $P_2(x)$ of degree ≤ 2 such that $P_2(x_i) = y_i, i = 0,1,2.$

Solution: Following the argument for the two point case, we plan to find the basis $l_0(x), l_1(x), l_2(x)$

We only construct the basis function $l_0(x)$ associate to the point x_0. Since from Equation (1.1), x_1, x_2 are zeros of $l_0(x)$, it should be of the form

Therefore; $l_0(x) = \dfrac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)},$

Similarly,

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_2-x_0)},$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)},$$

Hence the interpolation polynomial is as follows

$$P_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x).$$

Example3: Use global interpolation by one polynomial and piecewise polynomial interpolation with quadratic for the following nodes.

x_k	0	1	2	4	5
$f(x_k)$	0	16	48	88	0

Solution: Global interpolation by one polynomial: $P(x) = \sum_{k=0}^{4} L_{4,k}(x)\ f(x_k)$

$$P(x) = \frac{(x-1)(x-2)(x-4)(x-5)}{(0-1)(0-2)(0-4)(0-5)}(0) + \frac{(x-0)(x-2)(x-4)(x-5)}{(1-0)(1-2)(1-4)(1-5)}(16)$$

$$+ \frac{(x-0)(x-1)(x-4)(x-5)}{(2-0)(2-1)(2-4)(2-5)}(48) + \frac{(x-0)(x-1)(x-2)(x-5)}{(4-0)(4-1)(4-3)(4-5)}(88) + 0$$

Piecewise polynomial interpolation with quadratic

$$P(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)}(0) + \frac{(x-0)(x-2)}{(1-0)(1-2)}(16) + \frac{(x-0)(x-1)}{(2-0)(2-1)}(48); \qquad 0 \le x \le 2$$

$$P(x) = \frac{(x-4)(x-5)}{(2-4)(2-5)}(48) + \frac{(x-2)(x-5)}{(4-2)(4-5)}(88) + \frac{(x-2)(x-4)}{(5-2)(5-4)}(0); \qquad 2 \le x \le 5$$

4.1.2 Newton's divided differences method

To illustrate this method, linear and quadratic interpolation is presented first. Then, the general form of Newton's divided difference polynomial method is presented

In the preceding section, we discussed the problem of interpolating a function by a polynomial. We return to that problem now. Let f be a function whose values are known or computable at a set of points (nodes) $x_0, x_1, \cdots, x_n$. We assume in this section that these points are distinct, but they need NOT be ordered on the real line. We know that there exists a unique polynomial $P_n(x)$ of degree at most n that interpolates $f(x)$ at $n+1$ nodes:

$$P_n(x_i) = f(x_i), 0 \le i \le n.$$

Of course, the polynomial $P_n(x)$ can be constructed as a linear combination of the basis polynomial $1, x, x^2, \cdots, x^n$. As discussed in the previous section, this basis is not recommended. Another practical choice may be the Lagrange form of the interpolation polynomial, which as we have known can be easily derived. But, here, we prefer to use a basis appropriate to the Newton form of the interpolating polynomial:

Our task is to specify the coefficients $c_i, i = 0,1,\cdots,n$ by using the following data

x	x_0	x_1	x_2	$\cdots$	x_n	
$f(x)$	$f(x_0)$	$f(x_1)$	$f(x_2)$	$\cdots$	$f(x_n)$	

To this end, we first introduce the concept of the divided difference. Given x_i and the value $f(x_i)$, the zeroth order divided difference, is $f[x_i] = f(x_i)$.

Given x_i, x_j and values $f(x_i)$, $f(x_j)$, the first order divided difference is: $f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$.

Given $x_0, x_1, \cdots, x_n$, and values $f(x_0), f(x_1), \cdots, f(x_n)$, the high order divided difference is defined recursively: $f[x_0, x_1, x_2, \cdots, x_n] = \frac{f[x_0, x_1, \cdots, x_{n-1}] - f[x_1, x_2, \cdots, x_n]}{x_0 - x_n}$.

Example 1.

$$f[x_0, x_1] = \frac{f(x_0) - f(x_1)}{x_0 - x_1}$$

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} = \frac{\frac{f(x_0) - f(x_1)}{x_0 - x_1} - \frac{f(x_1) - f(x_2)}{x_1 - x_2}}{x_0 - x_2}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_0, x_1, x_2] - f[x_1, x_2, x_3]}{x_0 - x_3} = \frac{\frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} - \frac{f[x_1, x_2] - f[x_2, x_3]}{x_1 - x_3}}{x_0 - x_3}$$

The Newton Form of the interpolation polynomial. By the definition of the divided difference, for any x, we have $f[x, x_0] = \frac{f(x) - f(x_0)}{x - x_0}$.

Therefore, we have $f(x) = f(x_0) + f[x, x_0](x - x_0)$,

Using the definition of the third order divided difference, we have $f[x, x_0, x_1, x_2] = \frac{f[x, x_0, x_1] - f[x_0, x_1, x_2]}{x - x_2}$

$$f(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x, x_0, x_1, x_2, x_3](x - x_0)(x - x_2)(x - x_3)$$

.

This can now be solved for the term on the left side of involving x and then used to substitute into above Repeating the above argument gives

$$f(x) = f(x_0) + f[x_0, x_1](x - x_0) + \cdots + f[x_0, \cdots, x_n](x - x_0) \cdots (x - x_{n-1})$$

.

We take the *Newton form of the interpolation polynomial* as

$$N_n(x) = f(x_0) + f[x_0, x_1](x - x_0) + \cdots + f[x_0, \cdots, x_n](x - x_0) \cdots (x - x_{n-1})$$

.

Example 2

A robot arm with a rapid laser scanner is doing a quick quality check on holes drilled in a $15'' \times 10''$ rectangular plate. The centers of the holes in the plate describe the path the arm needs to take, and the hole centers are located on a Cartesian coordinate system (with the origin at the bottom left corner of the plate) given by the specifications in Table 3.2

Table 3.2 The coordinates of the holes on the plate.

x (in.)	2.00	4.25	5.25	7.81	9.20	10.60
y (in.)	7.2	7.1	6.0	5.0	3.5	5.0

Find the path traversed through the six points using Newton's divided difference method of interpolation and a fifth order polynomial.

Solution: For a fifth order polynomial, the value of y is given by

$$y(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$
$$+ b_4(x - x_0)(x - x_1)(x - x_2)(x - x_3) + b_5(x - x_0)(x - x_1)(x - x_2)(x - x_3)(x - x_4)$$

Using the six points, $x_0 = 2.00, \quad y(x_0) = 7.2$

$$x_1 = 4.25, \quad y(x_1) = 7.1$$
$$x_2 = 5.25, \quad y(x_2) = 6.0$$
$$x_3 = 7.81, \quad y(x_3) = 5.0$$
$$x_4 = 9.20, \quad y(x_4) = 3.5$$
$$x_5 = 10.60, \quad y(x_5) = 5.0$$

gives

$$b_0 = y[x_0] = y(x_0) = 7.2$$
$$b_1 = y[x_1, x_0]$$
$$= \frac{y(x_1) - y(x_0)}{x_1 - x_0} = \frac{7.1 - 7.2}{4.25 - 2.00}$$
$$= -0.044444$$
$$b_2 = y[x_2, x_1, x_0]$$
$$= \frac{y[x_2, x_1] - y[x_1, x_0]}{x_2 - x_0}$$
$$y[x_2, x_1] = \frac{y(x_2) - y(x_1)}{x_2 - x_1}$$
$$= \frac{6.0 - 7.1}{5.25 - 4.25} = -1.1$$
$$y[x_1, x_0] = -0.044444$$

$$b_2 = \frac{y[x_2, x_1] - y[x_1, x_0]}{x_2 - x_0}$$
$$= \frac{-1.1 + 0.044444}{5.25 - 2.00} = -0.32479$$
$$b_3 = y[x_3, x_2, x_1, x_0]$$
$$= \frac{y[x_3, x_2, x_1] - y[x_2, x_1, x_0]}{x_3 - x_0}$$
$$y[x_3, x_2, x_1] = \frac{y[x_3, x_2] - y[x_2, x_1]}{x_3 - x_1}$$
$$y[x_3, x_2] = \frac{y(x_3) - y(x_2)}{x_3 - x_2} = \frac{5.0 - 6.0}{7.81 - 5.25}$$
$$= -0.39063$$

$$y[x_2,x_1]=\frac{y(x_2)-y(x_1)}{x_2-x_1}=\frac{6.0-7.1}{5.25-4.25}$$
$$=-1.1$$
$$y[x_3,x_2,x_1]=\frac{y[x_3,x_2]-y[x_2,x_1]}{x_3-x_1}=\frac{-0.39063+1.1}{7.81-4.25}$$
$$=0.19926$$
$$y[x_2,x_1,x_0]=-0.32479$$

$$b_3=y[x_3,x_2,x_1,x_0]$$
$$=\frac{y[x_3,x_2,x_1]-y[x_2,x_1,x_0]}{x_3-x_0}=\frac{0.19926+0.32479}{7.81-2.00}$$
$$=0.090198$$

$$b_4=y[x_4,x_3,x_2,x_1,x_0]$$
$$=\frac{y[x_4,x_3,x_2,x_1]-y[x_3,x_2,x_1,x_0]}{x_4-x_0}$$
$$y[x_4,x_3,x_2,x_1]=\frac{y[x_4,x_3,x_2]-y[x_3,x_2,x_1]}{x_4-x_1}$$
$$y[x_4,x_3,x_2]=\frac{y[x_4,x_3]-y[x_3,x_2]}{x_4-x_2}$$
$$y[x_4,x_3]=\frac{y(x_4)-y(x_3)}{x_4-x_3}=\frac{3.5-5.0}{9.20-7.81}$$
$$=-1.0791$$
$$y[x_3,x_2]=-0.39063$$
$$y[x_4,x_3,x_2]=\frac{y[x_4,x_3]-y[x_3,x_2]}{x_4-x_2}=\frac{-1.0791+0.39063}{9.20-5.25}$$
$$=-0.17431$$
$$y[x_3,x_2,x_1]=0.19926$$
$$y[x_4,x_3,x_2,x_1]=\frac{y[x_4,x_3,x_2]-y[x_3,x_2,x_1]}{x_4-x_1}=\frac{-0.17431-0.19926}{9.20-4.25}$$
$$=-0.075469$$
$$y[x_3,x_2,x_1,x_0]=0.090198$$

$$b_4=\frac{y[x_4,x_3,x_2,x_1]-y[x_3,x_2,x_1,x_0]}{x_4-x_0}=\frac{-0.075469-0.090198}{9.20-2.00}$$
$$=-0.023009$$

$$b_5=y[x_5,x_4,x_3,x_2,x_1,x_0]$$

$$= \frac{y[x_5, x_4, x_3, x_2, x_1] - y[x_4, x_3, x_2, x_1, x_0]}{x_5 - x_0}$$

$$y[x_5, x_4, x_3, x_2, x_1] = \frac{y[x_5, x_4, x_3, x_2] - y[x_4, x_3, x_2, x_1]}{x_5 - x_1}$$

$$y[x_5, x_4, x_3, x_2] = \frac{y[x_5, x_4, x_3] - y[x_4, x_3, x_2]}{x_5 - x_2}$$

$$y[x_5, x_4, x_3] = \frac{y[x_5, x_4] - y[x_4, x_3]}{x_5 - x_3}$$

$$y[x_5, x_4] = \frac{y(x_5) - y(x_4)}{x_5 - x_4} = \frac{5.0 - 3.5}{10.60 - 9.20}$$

$$= 1.0714$$

$$y[x_4, x_3] = -1.0791$$

$$y[x_5, x_4, x_3] = \frac{y[x_5, x_4] - y[x_4, x_3]}{x_5 - x_3} = \frac{1.0714 + 1.0791}{10.60 - 7.81}$$

$$= 0.77081$$

$$y[x_4, x_3, x_2] = -0.17431$$

$$y[x_5, x_4, x_3, x_2] = \frac{y[x_5, x_4, x_3] - y[x_4, x_3, x_2]}{x_5 - x_2}$$

$$= \frac{0.77081 + 0.17431}{10.60 - 5.25} = 0.17666$$

$$y[x_4, x_3, x_2, x_1] = -0.075469$$

$$y[x_5, x_4, x_3, x_2, x_1] = \frac{y[x_5, x_4, x_3, x_2] - y[x_4, x_3, x_2, x_1]}{x_5 - x_1}$$

$$= \frac{0.17666 + 0.075469}{10.60 - 4.25} = 0.039705$$

$$y[x_4, x_3, x_2, x_1, x_0] = -0.023009$$

$$b_5 = \frac{y[x_5, x_4, x_3, x_2, x_1] - y[x_4, x_3, x_2, x_1, x_0]}{x_5 - x_0}$$

$$= \frac{0.039705 + 0.023009}{10.60 - 2.00} = 0.0072923$$

Hence

$$\begin{aligned} y(x) &= b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2) \\ &\quad + b_4(x - x_0)(x - x_1)(x - x_2)(x - x_3) \\ &\quad + b_5(x - x_0)(x - x_1)(x - x_2)(x - x_3)(x - x_4) \\ &= 7.2 - 0.04444(x - 2) - 0.32479(x - 2)(x - 4.25) \\ &\quad + 0.090198(x - 2)(x - 4.25)(x - 5.25) \\ &\quad - 0.023009(x - 2)(x - 4.25)(x - 5.25)(x - 7.81) \\ &\quad + 0.0072923(x - 2)(x - 4.25)(x - 5.25)(x - 7.81)(x - 9.2) \end{aligned}$$

Expanding this formula, we get

$$y(x) = -30.898 + 41.344x - 15.855x^2 + 2.7862x^3 - 0.23091x^4 + 0.0072923x^5, \quad 2 \le x \le 10.6$$

- **linear Interpolation**

Given (x_0, y_0) and (x_1, y_1), fit a linear interpolant through the data. Noting $y = f(x)$ and $y_1 = f(x_1)$, assume the linear interpolant $f_1(x)$ is given by (Figure 2)

$$f_1(x) = b_0 + b_1(x - x_0)$$

Since at $x = x_0$,

$$f_1(x_0) = f(x_0) = b_0 + b_1(x_0 - x_0) = b_0$$

and at $x = x_1$,

$$f_1(x_1) = f(x_1) = b_0 + b_1(x_1 - x_0)$$
$$= f(x_0) + b_1(x_1 - x_0)$$

Giving $b_1 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}$

So $b_0 = f(x_0)$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Giving the linear interpolant as

$$f_1(x) = b_0 + b_1(x - x_0)$$

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

Example 1: The upward velocity of a rocket is given as a function of time in Figure below

Table3.3: Velocity as a function of time.

t (s)	0	10	15	20	22.5	30
$v(t)$ (m/s)	0	227.04	362.78	517.35	602.97	901.67

.

Determine the value of the velocity at $t = 16$ seconds using first order polynomial interpolation by Newton's divided difference polynomial method.

Solution: For linear interpolation, the velocity is given by

$$v(t) = b_0 + b_1(t - t_0)$$

Since we want to find the velocity at $t = 16$, and we are using a first order polynomial, we need to choose the two data points that are closest to $t = 16$ that also bracket $t = 16$ to evaluate it. The two points are $t = 15$ and $t = 20$.

Then

$$t_0 = 15, \; v(t_0) = 362.78$$

$$t_1 = 20, \; v(t_1) = 517.35$$

Gives $b_0 = v(t_0)$

$$= 362.78$$

$$b_1 = \frac{v(t_1) - v(t_0)}{t_1 - t_0}$$
$$= \frac{517.35 - 362.78}{20 - 15}$$
$$= 30.914$$

Hence $v(t) = b_0 + b_1(t - t_0)$

$= 362.78 + 30.914(t - 15), \quad 15 \le t \le 20$

At $t = 16,\ v(16) = 362.78 + 30.914(16 - 15)$

$= 393.69$ m/s

If we expand $v(t) = 362.78 + 30.914(t - 15),\ 15 \le t \le 20$

we get $v(t) = -100.93 + 30.914t, \quad 15 \le t \le 20$

and this is the same expression as obtained in the direct method.

- **Quadratic Interpolation**

Given $(x_0, y_0), (x_1, y_1),$ and $(x_2, y_2),$ fit a quadratic interpolant through the data. Noting $y = f(x),\ y_0 = f(x_0),\ y_1 = f(x_1),$ and $y_2 = f(x_2),$ assume the quadratic interpolant $f_2(x)$ is given by $f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$

At $x = x_0,\ f_2(x_0) = f(x_0) = b_0 + b_1(x_0 - x_0) + b_2(x_0 - x_0)(x_0 - x_1)$

$= b_0$

$b_0 = f(x_0)$

At $x = x_1\ f_2(x_1) = f(x_1) = b_0 + b_1(x_1 - x_0) + b_2(x_1 - x_0)(x_1 - x_1)$

$f(x_1) = f(x_0) + b_1(x_1 - x_0)$

Giving $b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$

At $x = x_2\ f_2(x_2) = f(x_2) = b_0 + b_1(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$

$$f(x_2) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$$

Giving $b_2 = \dfrac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$

Hence the quadratic interpolant is given by

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$
$$= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}(x - x_0)(x - x_1)$$

Example. Compute a divided difference table for these function values.

x	3	1	5	6
$f(x)$	1	-3	2	4

4.1.3 Neville's Method

However, the best way to calculate large interpolation polynomials is to use *Neville's Algorithm*

- Neville's algorithm is better because it builds up the interpolation polynomial in a recursive fashion
- The employed recursion cuts down the total number of operations required and helps reduce concerns about rounding error

Theorem: Let f be defined at $x_0, x_1, \cdots, x_k$, and let x_j and x_i be two distinct numbers in this set. Then $$P(x) = \frac{(x-x_j)P_{0,1,\cdots,j-1,j+1,\cdots,k}(x) - (x-x_i)P_{0,1,\cdots,i-1,i+1,\cdots,k}(x)}{(x_i - x_j)}$$ describes the kth Lagrange polynomial that interpolates f at the $k+1$ points $x_0, x_1, \cdots, x_k$.

$x_0 \quad P_0 = Q_{0,0}$

$x_1 \quad P_1 = Q_{1,0} \quad P_{0,1} = Q_{1,1}$

$x_2 \quad P_2 = Q_{2,0} \quad P_{1,2} = Q_{2,1} \quad P_{0,1,2} = Q_{2,2}$

$x_3 \quad P_3 = Q_{3,0} \quad P_{2,3} = Q_{3,1} \quad P_{1,2,3} = Q_{3,2} \quad P_{0,1,2,3} = Q_{3,3}$

$x_4 \quad P_4 = Q_{4,0} \quad P_{3,4} = Q_{4,1} \quad P_{2,3,4} = Q_{4,2} \quad P_{1,2,3,4} = Q_{4,3} \quad P_{0,1,2,3,4} = Q_{4,4}$

This procedure is called **Neville's Method**. Neville's Method uses the nodes $x_{i-k}, x_{i-k+1}, \cdots, x_i$. Let $P_{i-k,i-k+1,\cdots,i-1,i}(x) = Q_{i,k}(x)$, we have

$$Q_{i,k}(x) = \frac{(x-x_i)}{(x_{i-k}-x_i)} Q_{i-1,k-1}(x) + \frac{(x-x_{i-k})}{(x_i - x_{i-k})} Q_{i,k-1}(x)$$

$$= \frac{(x-x_{i-k})Q_{i,k-1}(x) - (x-x_i)Q_{i-1,k-1}(x)}{(x_i - x_{i-k})}$$

$$k = 1, 2, \cdots; \quad i = k, k+1, \cdots.$$

Example 1: The following table lists the values of $f(x) = \ln x$ accurate to the places given. Using Neville's method to approximate $f(2.1) = \ln(2.1)$, and find the error bound

Table 3.4

i	x_i	$\ln x_i$
0	2.0	0.6931
1	2.2	0.7885
2	2.3	0.8329

Using Neville's method, $Q_{i,j}$ $(0 \le j \le i)$, the interpolating polynomial of degree j on the $(j+1)$ numbers $x_{i-j}, x_{i-j+1}, \cdots, x_{i-1}, x_i$ are listed in Table 3.6:

Table 3.4

$i \qquad x_i \qquad Q_{i,0} \qquad Q_{i,1} \qquad Q_{i,2}$

---

0 2.0 0.6932
1 2.2 0.7885 0.7410
2 2.3 0.8329 0.7441 0.7420

Thus, $P_2(2.1) = Q_{2,2} = 0.7420$. Since $f(2.1) = \ln(2.1) = 0.7419$ to four decimal places, the absolute error is

$$|f(2.1) - P_2(2.1)| = |0.7419 - 0.7420| = 10^{-4}.$$

Example 2: Using Neville's Method to obtain the approximation $f(1.5)$ if $x_0 = 1$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$, $x_4 = 2.2$, and $Q_{0,0} = f(x_0) = f(1.0) = 0.7651977$, $Q_{1,0} = f(x_1) = f(1.3) = 0.6200860$, $Q_{2,0} = f(x_2) = f(1.6) = 0.4554022$, $Q_{3,0} = f(x_3) = f(1.9) = 0.2818186$, $Q_{4,0} = f(x_4) = f(2.2) = 0.1103623$.

These are the five polynomials of degree zero (constants) that approximate $f(1.5)$.

Calculating the first-degree approximation $Q_{1,1}(1.5)$ gives

$$Q_{1,1}(1.5) = \frac{(x - x_0)Q_{1,0}(x) - (x - x_1)Q_{0,0}(x)}{(x_1 - x_0)}$$

$$= \frac{(1.5 - 1.0)Q_{1,0}(1.5) - (1.5 - 1.3)Q_{0,0}(1.5)}{(13 - 1.0)}$$

$$= \frac{0.5 \times 0.6200860 - 0.2 \times 0.7651977}{0.3} = 0.5233449$$

Similarly,

$$Q_{2,1}(1.5) = \frac{(x - x_1)Q_{2,0}(x) - (x - x_2)Q_{1,0}(x)}{(x_2 - x_1)}$$

$$= \frac{(1.5 - 1.3)Q_{2,0}(1.5) - (1.5 - 1.6)Q_{1,0}(1.5)}{(1.6 - 1.3)}$$

$$= \frac{0.5 \times 0.4554022 - 0.2 \times 0.6200860}{0.3} = 0.5102968$$

$$Q_{3,1}(1.5) = \frac{(x - x_2)Q_{3,0}(x) - (x - x_3)Q_{2,0}(x)}{(x_3 - x_2)} = 0.5132636$$

$$Q_{4,1}(1.5) = \frac{(x - x_3)Q_{4,0}(x) - (x - x_4)Q_{3,0}(x)}{(x_4 - x_3)} = 0.5104270$$

The best linear approximation is expected to be $Q_{2,1}$, since 1.5 is between $x_1 = 1.3$ and $x_2 = 1.6$.

In a similar manner, approximations using higher-degree polynomials are given by

$$Q_{2,2}(1.5) = \frac{(1.5 - x_0)Q_{2,1}(1.5) - (1.5 - x_2)Q_{1,1}(1.5)}{(x_2 - x_0)} = 0.5124715$$

$$Q_{3,2}(1.5) = \frac{(1.5 - x_1)Q_{3,1}(1.5) - (1.5 - x_3)Q_{2,1}(1.5)}{(x_3 - x_1)} = 0.5112857$$

$$Q_{4,2}(1.5) = \frac{(1.5 - x_2)Q_{4,1}(1.5) - (1.5 - x_4)Q_{3,1}(1.5)}{(x_4 - x_2)} = 0.5137361.$$

The higher-degree approximations are generated in a similar manner and are shown in Table

Table 3.5

1.0	0.7651977				
1.3	0.6200860	0.5233449			
1.6	0.4554022	0.5102968	0.5124715		
1.9	0.2818186	0.5132634	0.5112857	0.5118127	
2.2	0.2203623	0.5104270	0.5137361	0.5118302	0.5118200

If the latest approximation, $Q_{4,4}$ is not sufficiently accurate, another node, x_5, can be selected, and another row added to the table:

x_5 $Q_{5.0}$ $Q_{5.1}$ $Q_{5.2}$ $Q_{5.3}$ $Q_{5.4}$ $Q_{5.5}$.

Advantages and Disadvantages of Interpolating Polynomials

- They are "perfect" in the curve fitting sense because they pass through all the data points.
- They are easy to use for evaluation, differentiation and integration.
- Extreme caution must be used to extrapolate for values outside the intervals of the independent variable for which data has been gathered.
- They tend to oscillate severely near the endpoints of the data range. This leads to errors when interpolating near the endpoints and extreme error when estimating the derivative near the endpoints.
- The coefficients can be extremely sensitive to small changes in the data.

4.2 Interpolation with Cubic Spline

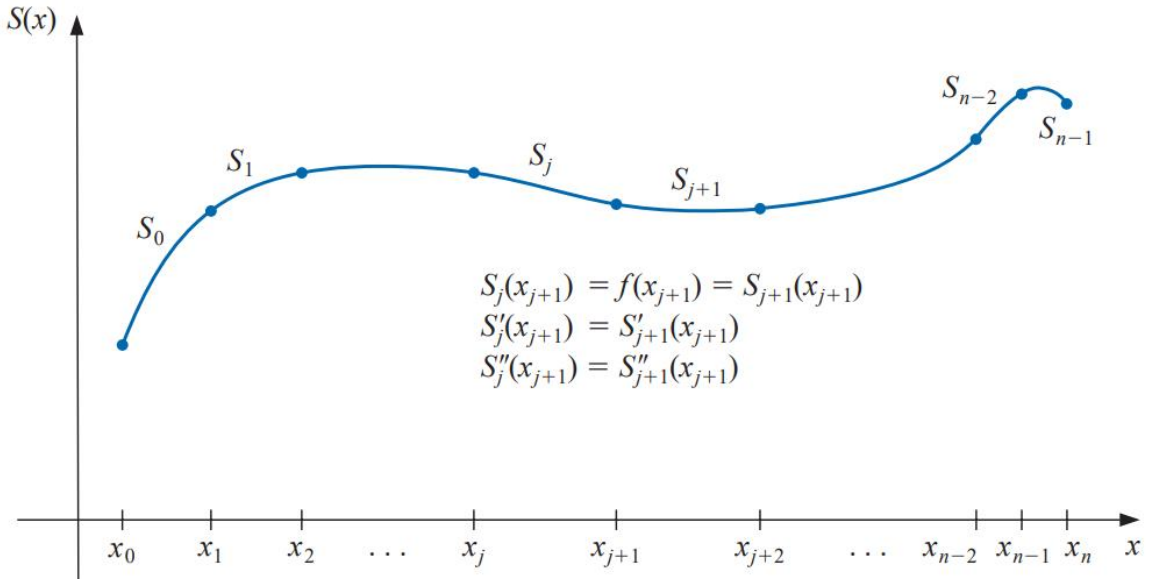
The goal of cubic spline interpolation is to get an interpolation formula that is continuous in both the first and second derivatives, both within the intervals and at the interpolating nodes. This will give us a smoother interpolating function. In general, if the function you want to approximate is smooth, then cubic splines will do better than piecewise linear interpolation.

Cubic Spline Interpolation

The previous sections concerned the approximation of arbitrary functions on closed intervals using a single polynomial. However, high-degree polynomials can oscillate erratically, that is, a minor fluctuation over a small portion of the interval can induce large fluctuations over the entire range.

An alternative approach is to divide the approximation interval into a collection of subintervals and construct a (generally) different approximating polynomial on each subinterval. This is called **piecewise-polynomial approximation.**

The most common piecewise-polynomial approximation uses cubic polynomials between each successive pair of nodes and is called cubic spline interpolation.



Given a function f defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \cdots < x_n = b$, a **cubic spline interpolant** S for f is a function that satisfies the following conditions:

(a) $S(x)$ is a cubic polynomial, denoted $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, 1, \ldots, n-1$;

(b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each $j = 0, 1, \ldots, n-1$;

(c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$; (*Implied by* **(b)**.)

(d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$;

(e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$;

(f) One of the following sets of boundary conditions is satisfied:

(i) $S''(x_0) = S''(x_n) = 0$ (**natural** *(or free)* **boundary**);

(ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (**clamped boundary**). ■

Example 1 Construct a natural cubic spline that passes through the points (1, 2), (2, 3), and (3, 5).

Solution This spline consists of two cubics. The first for the interval [1, 2], denoted

$S_0(x) = a_0 + b_0(x - 1) + c_0(x - 1)^2 + d_0(x - 1)^3$, and

the other for [2, 3], denoted

$S_1(x) = a_1 + b_1(x - 2) + c_1(x - 2)^2 + d_1(x - 2)^3$.

There are 8 constants to be determined, which requires 8 conditions. Four conditions come from the fact that the splines must agree with the data at the nodes. Hence

$2 = f(1) = a_0$, $3 = f(2) = a_0 + b_0 + c_0 + d_0$, $3 = f(2) = a_1$, and

$5 = f(3) = a_1 + b_1 + c_1 + d_1$.

Two more come from the fact that $S'_0(2) = S'_1(2)$ and $S''_0(2) = S''_1(2)$. These are

$S'_0(2) = S'_1(2)$; $b_0 + 2c_0 + 3d_0 = b_1$ and $S''_0(2) = S''_1(2)$. : $2c_0 + 6d_0 = 2c_1$

The final two come from the natural boundary conditions:

$S_0''(1) = 0$: $2c_0 = 0$ and $S_1''(3) = 0$: $2c_1 + 6d_1 = 0$.

Solving this system of equations gives the spline

$$S(x) = \begin{cases} 2 + \frac{3}{4}(x-1) + 14(x-1)^3, \text{for } x \in [1,2] \\ 3 + \frac{3}{2}(x-2) + \frac{3}{4}(x-2)^2 - \frac{1}{4}(x-2)^3, \text{for } x \in [2,3] \end{cases}$$

4.3 Least-Squares Curve Fitting

Linear Regression

We'll create equation of the **least squares line**, which is also called **best-fit line** or **regression line**. The line is passing in between our points while the sum of the squares of the vertical distances from the data points to the line is as small as possible. The picture to the left shows the least square line passing in between our points, and the distances d_1, d_2, d_3, d_4, and d_5. The equation of the least square line is found by minimizing the sum:

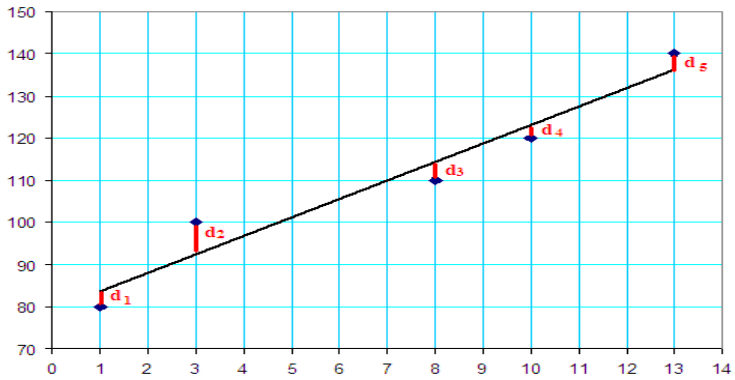
$(d_1)^2 + (d_2)^2 + (d_3)^2 + (d_4)^2 + (d_5)^2$

The procedure will be omitted in this paper. The final result of the minimization are formulas that let us calculate coefficients *a* and *b* for equation of the least square line y= ax+b. This equation may be used for the prediction of sales.

FORMULAS FOR COEFFICIENTS OF THE LEAST SQUARES LINE:

$$a = \frac{n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)}{n\left(\sum x^2\right) - \left(\sum x\right)^2} \qquad b = \frac{\sum y - a \cdot \left(\sum x\right)}{n}$$

Example: Find the linear representation of the following data representation



In this formula *n* stands for number of observed cases. In our case that is 5 employees. So: $n = 5$

Symbol $\sum$ is called "sigma" and stands for the "sum of all". In our case:

$\sum x = 1 + 3 + 8 + 10 + 13 = 35$ *(sum of all x values)*

$\sum y = 80 + 100 + 110 + 110 + 120 + 140 = 550$ *(sum of all y values)*

$\sum x^2 = 1^2 + 3^2 + 8^2 + 10^2 + 13^2 = 343$ *(sum of squares of x values)*

$\sum xy = 1 \cdot 80 + 3 \cdot 100 + 8 \cdot 110 + 10 \cdot 120 + 13 \cdot 140 = 4280$ *(sum of xy products)*

We are now ready to use results of our calculations in the formula:

$$a = \frac{5 \cdot 4280 - 35 \cdot 550}{5 \cdot 343 - 35^2} = \frac{2150}{490} \approx 4.388 \qquad b = \frac{550 - 4.388 \cdot 35}{5} = \frac{396.42}{5} \approx 79.284$$

That means that the equation of the least square line is $y = 4.388x + 79.284$

Conclusion:

We created the formula $y = 4.388x + 79.284$ that may be used to predict sales in thousands for a future employee. For example, formula may predict that an employee with x=15 years of experience will generate:

$y = 4.388 \cdot 15 + 79.284 \approx 145$ thousands in sales per year.

4.4 Least Squares Fitting--Polynomial

A commonly used linear form is a polynomial. If the degree of the polynomial is m − 1, We have polynomial

$$y = a_0 + a_1 x + \ldots + a_k x^k,$$

the residual is given by

$$R^2 \equiv \sum_{i=1}^{n} \left[y_i - \left(a_0 + a_1 x_i + \ldots + a_k x_i^k \right) \right]^2.$$

The partial derivatives (again dropping superscripts) are

$$\frac{\partial (R^2)}{\partial a_0} = -2 \sum_{i=1}^{n} \left[y - \left(a_0 + a_1 x + \ldots + a_k x^k \right) \right] = 0$$

$$\frac{\partial (R^2)}{\partial a_1} = -2 \sum_{i=1}^{n} \left[y - \left(a_0 + a_1 x + \ldots + a_k x^k \right) \right] x = 0$$

$$\frac{\partial (R^2)}{\partial a_k} = -2 \sum_{i=1}^{n} \left[y - \left(a_0 + a_1 x + \ldots + a_k x^k \right) \right] x^k = 0.$$

These lead to the equations

$$a_0 n + a_1 \sum_{i=1}^{n} x_i + \ldots + a_k \sum_{i=1}^{n} x_i^k = \sum_{i=1}^{n} y_i$$

$$a_0 \sum_{i=1}^{n} x_i + a_1 \sum_{i=1}^{n} x_i^2 + \ldots + a_k \sum_{i=1}^{n} x_i^{k+1} = \sum_{i=1}^{n} x_i y_i$$

$$a_0 \sum_{i=1}^{n} x_i^k + a_1 \sum_{i=1}^{n} x_i^{k+1} + \ldots + a_k \sum_{i=1}^{n} x_i^{2k} = \sum_{i=1}^{n} x_i^k y_i$$

or, in matrix form

$$\begin{bmatrix} n & \sum_{i=1}^{n} x_i & \cdots & \sum_{i=1}^{n} x_i^k \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 & \cdots & \sum_{i=1}^{n} x_i^{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^{n} x_i^k & \sum_{i=1}^{n} x_i^{k+1} & \cdots & \sum_{i=1}^{n} x_i^{2k} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \\ \vdots \\ \sum_{i=1}^{n} x_i^k y_i \end{bmatrix}.$$